Digital forensics concepts, investigations and digital evidence

Dr Olga Angelopoulou

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Lecture Objectives

- Develop an understanding of the core principles of digital forensics
- Identify the extent and role of understanding cybercrime for digital forensic investigations
- Familiarise with the concept of digital evidence
- Develop a critical understanding of the types of digital investigations

Cybercrimes

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Cybercrime & Cyberattack Timeline (2021–2025)

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2021

May 7
Colonial Pipeline ransomware attack (USA)

- Attack on major fuel pipeline; ransom paid after disruption

July 2
Kaseya VSA ransomware

- Vulnerability exploited in MSP software affecting thousands worldwide [4]

2022–2023

May 14
Health Service Executive (HSE) ransomware attack

- Major ransomware attack severely impacted national health service

July 2
Kaseja VSA ransomware **attack** (global) [3]

- Vulnerability exploited in MSP software affecting thousands worldwide

2024

June–August
Kadokawa & Niconico cyberattack

- Ransomware attack on major digital platforms; user data leaked

2025

June 11
Paraguay ransomware atttack [6]

- Government targeted; ransomware group demanded **$7.4M** [5]

Kido International ransomware (UK/international) [6,7]

- Hit multinattonal education provider: data of children & staff exposed
- Ransomware attempts projected at **11,000**/day by late 2025

Other Noteworthy Patterns (2021–2025)

- Cybercrime attacks quickly rose in 2021 (+125% from 2020), continuing to grow through 2022 [6] -8, [9]
- Critical infrastructure & essential services are now **frequent targets**

References

[1] **Darktrase. Q1 SPI report reveals** record surge and diversification in **global cyber attancs**, 2021

[2] **BSC News.** HSE cyber atack What is ransomware and who was the high health service targeted 2022

[3] **SecurityWeek.** Kaseye sem up +0 1,500 businesses hit by tm Ransom-wark Adta 21 2021

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Source: Public reporting (2021–2025). Visualisation generated with ChatGPT (OpenAI).

What is Crime?

1.[uncountable] activities that involve breaking the law

2.[countable] **crime (against somebody)** an illegal act or activity that can be punished by law

Oxford Learner's Dictionaries

- There are two types of laws:
  - Natural laws
    - protect against harm to persons (e.g. murder, rape, assault) or property (theft, larceny, robbery)
    - form the basis of common law systems.
  - Statutes are enacted by legislatures and reflect current cultural mores

Cybercrime

1. Cybercrime is an act that violates the law, which is perpetrated using information and communication technology (ICT) to either target networks, systems, data, websites and/or technology or facilitate a crime

2. Offences against the confidentiality, integrity and availability of computer information and systems.

Knows no physical or geographic boundaries

- Computer crime
- Network crime
- Computer related crime
- Computer-facilitated crime
- High tech crime
- Internet crime/ Online crime
- Information age crime

Types of Cybercrime

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Cyber-dependent

- any crime that can only be committed using computers, computer networks or other forms of information communication technology

Cyber-enabled

- traditional crimes facilitated by the Internet and digital technologies

Computers and crimes

Facilitator of crime

Enabler of crime

Promoter of crime

(Kirwan& Power, 2013)

	Contraband	Fruits of Crime	Instrumentality	Evidence
Hardware	Cloned mobile telephones, or hardware for intercepting communications	Stolen computers, or equipment purchased with stolen credit cards	Printer used to produce counterfeit banknotes, or scanner used to produce child pornography	Mobile phone may be evidence of parole violation even if it was not used to deal drugs
Information	Digital photographs or videos of child exploitation, or strong encryption in some countries	Valuable data stolen from computers such as bank account details	Programs used to break into computers and capture passwords	A personal diary on a computer describing details of a crime, or log files showing criminal activity

Searching and Seizing Computers and Obtaining Electronic Evidence in Criminal Investigations (2001)

Adversary Model of Cybercrime

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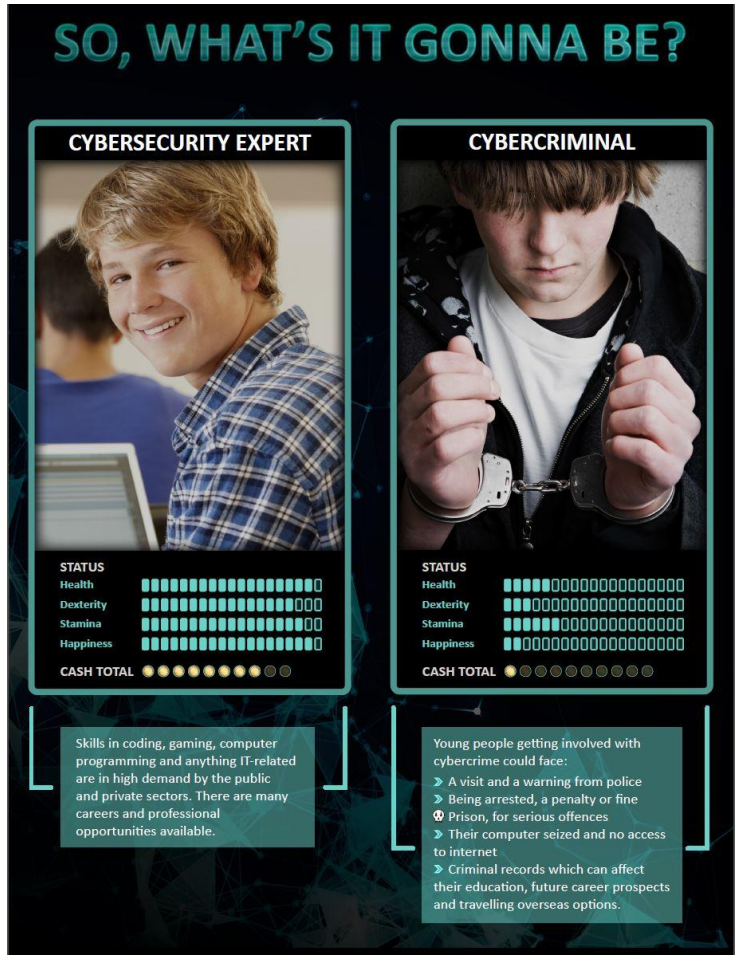
Resources	Capabilities	Intent
Motivation	Risk aversion	Access

- A framework for understanding attacker behaviour
- Helps identify capabilities, intent, and resources
- Supports risk assessment and defensive planning
- Widely used in security and threat modelling

Taxonomy of Cybercriminals

Category	Motivation	Skills
Novice	Looking for media attention	• Limited computer and programming skills. • Rely on toolkits to conduct their attacks. • Can cause extensive system damage as they don't understand how the attack works.
Cyber-punks	• Deliberately attack/ vandalise • Tend to brag about their exploits	• Some skills • Capable of writing their own software. • Understand the systems they are attacking.
Insiders	Disgruntled employees or ex-employees	• Computer literate • May be involved in technology-related jobs. • Aided by privileges they have or had been assigned as part of their job function.
Old-guard hackers	• no criminal intent • alarming disrespect for personal property	• High skills • Interested in the intellectual endeavour
Professional criminals	• Motivated by profit • Specialise in corporate espionage	• High skills, well-equipped, professional criminals • Highly motivated, highly trained • Guns for hire
Cyber-terrorists	Motivated by ideals	• High skills, well-equipped, professional criminals • Highly motivated, highly trained • Well funded • Political activists



Challenges in combating cybercrime

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Technical

- attribution
- vulnerabilities
- virtualised infrastructure

Legal

- transnational crime
- different legal systems between countries
- variations in national cybercrime laws
- differences in the rules of evidence and criminal procedure
- variations in the scope and geographic applicability of regional and multilateral cybercrime treaties
- differences in approaches to data protection and respect for human rights.

Ethical

- refraining from harming others, systems, and data, and respecting the rule of law and human rights

Operational

- cooperation with other countries

Types of Digital Investigations

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Criminal forensics

- Law enforcement focused - the investigation may be contracted
- Evidence often used in court proceedings
- Focus on forensically sound data extraction and report

Corporate and Workplace Investigations

- Internal organisational investigations
- Often involves HR and legal teams

Intelligence and Threat Investigations

- Proactive rather than reactive
- Often associated with crime
- Related to providing intelligence to help track, stop or identify criminal activity
- Supports risk assessment and strategic decisions

Incident response

- Responding to cyber security incidents
- Investigation focuses on identifying the entry point for such attacks
- Containment, eradication, and recovery

Civil Litigations

- Related to civil law
- Support for legal disputes between parties
- Functionally identical to its criminal counter part
- Often involves e-discovery processes
- eDiscovery has specific legal limitations and restrictions

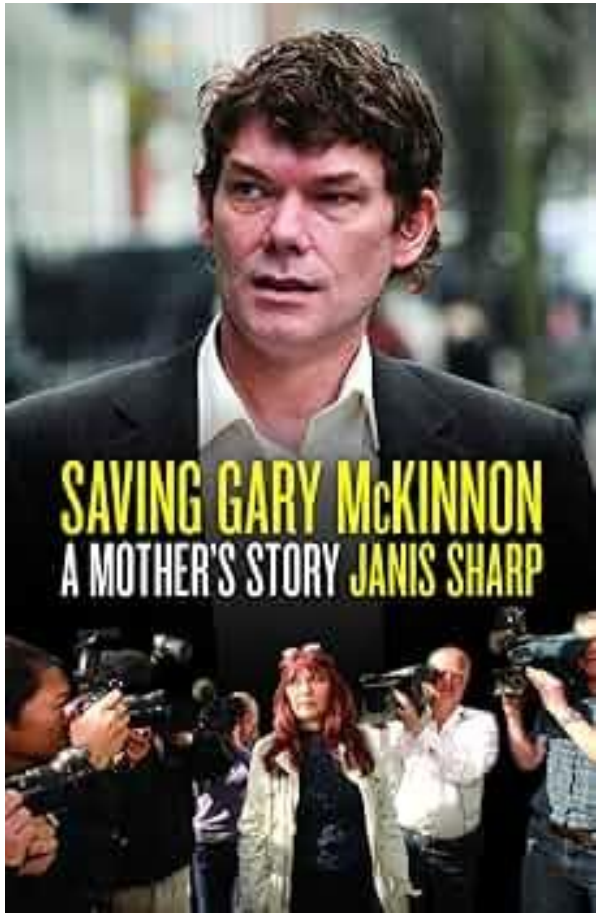
Investigation Goal

1. **Who** is responsible for the activity? - Attribution
2. **What** has the cybercriminal/ attacker done? - Determination and motive e.g. files accessed and mode of access, network traffic generated, user accounts attacked
3. **When** did the event happen? – Timeline
4. **Where** is the origin of the crime/ attack? - Location or identity of the cybercriminal's system/ operation
5. **Why** did it happen? – Purpose/ motive
6. **How** was the attack executed? – Reconstruction of the event

Discussion Point 1

The case of Gary McKinnon

Research on what went wrong that led to his arrest



> British computer hacker Gary McKinnon has been living under the threat of extradition and prosecution in the US for more than 10 years.

https://www.theguardian.com/world/2012/oct/16/gary-mckinnon-timeline-extradition

Digital Forensics Principles

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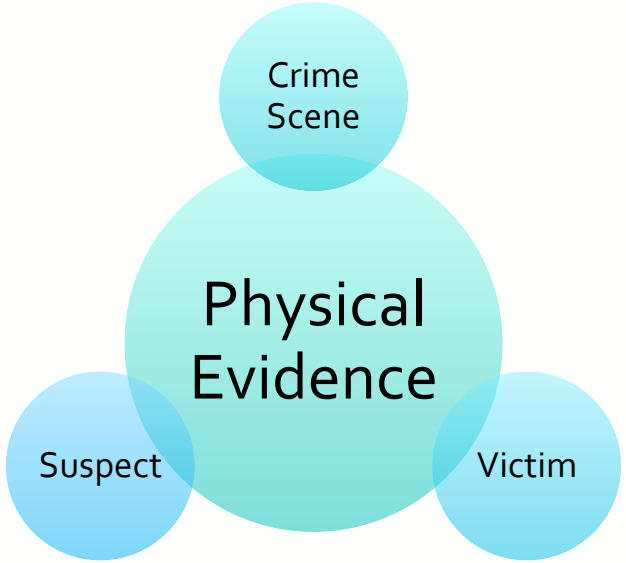
What is Forensic Science?

Forensics – Latin: meaning "of or before the forum" dating back to Roman times, when being charged with a crime would result in having to make speeches for (effectively) prosecution and defence in front of a group of individuals in the forum.

Forensic science is used to give insight to the chain of events that occurred during a crime.

1. Forensic scientists search and examine physical traces for establishing or excluding an association between someone suspected of committing a crime and the scene of the crime or victim.
2. Forensic scientists can appear for criminal matters or civil matters
3. Forensic scientists present their findings and opinions in written form either as formal statements of evidence or reports.



Locard's Exchange Principle

"Toute action de l'homme, et a fortiori, l'action violente qu'est un crime, ne peut pas se dérouler sans laisser quelque marque"

"Any action of an individual, and obviously, the violent action constituting a crime, cannot occur without leaving a mark.

What is admirable is the variety of these marks. Sometimes they will be prints, sometimes simple traces, and sometimes stains."

Dr Edmond Locard

Digital Forensics

the scientific analysis of a digital device to extract evidential value from it

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The science of **acquiring**, **retrieving**, **preserving**, and **presenting** data that has been processed electronically and stored on computer media.

(Schweitzer, 2003)

- The application of scientific methods to digital evidence
- Supports investigations, incident response, and legal proceedings
- Focuses on integrity, repeatability, and reliability

Locard's Exchange Principle in a digital context

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If you are connecting to or using a computer system, assorted changes take place on the devices used.

- Files are created, modified and destroyed
- Logs are written
- Actions are carried out
- Data in memory is changed
- Data is exchanged over a network

ACPO Guidelines

The **Principles of Computer-Based Electronic Evidence**

- **Principle 1: integrity** of the evidence must be provably maintained
- **Principle 2:** examiners should be **appropriately qualified**, and be able to **explain and/or document** their actions and findings
- **Principle 3:** maintain **contemporaneous notes**
- **Principle 4:** Many roles may be involved by one or more people. **One person** is assigned as in charge

Principle 1

- **No action** taken by law enforcement agencies or their agents **should change data** held on a computer or storage media which may subsequently be relied upon in court.

Principle 2

- In exceptional circumstances, where a person finds it necessary to access original data held on a computer or on storage media, that person must **be competent to do so and be able to give evidence** explaining the relevance and the implications of their actions.

Principle 3

- An **audit trail** or other record of all processes applied to computer based electronic evidence should be created and preserved. An independent third party should be able to examine those processes and achieve the same result.

Principle 4

- The **person in charge** of the investigation **has overall responsibility** for ensuring that the law and these principles are adhered to.

The Principles

- Protect evidence from challenge
- Ensure fairness in investigations
- Support credibility of forensic findings

Integrity of Evidence

- Preservation is critical
- Write blockers where appropriate
- Hashing used to verify integrity
- Minimise interaction with original evidence

Repeatability and Reproducibility

- Processes must be repeatable by another examiner
- Methods should be documented
- Tools should be validated where possible

Chain of Custody

- Documents who handled the evidence
- Records time, date, and actions taken
- Essential for legal admissibility

Proportionality and Ethics

- Only collect what is necessary
- Respect privacy and legalboundaries
- Consider professional codes of conduct

Chain of Custody

The Law of Evidence or else "paper trail"

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Demonstrate that the digital evidence presented is original and can stand in court – it has not been tampered with.

documenting everything

- Where it is **at all times** – Collection and transfer
- Who has it **at all times** - Sequence of control and custody, dates and times
- What has been done to it - Analysis and purpose

Standards and Other Initiatives

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For criminal cases, the UK Forensic Science Regulator (FSR) publishes codes of practice:

- All forensic laboratories in the UK are required to comply with
  - **ISO 17025:2005** (General requirements for the competence of testing and calibration laboratories)
  - **ISO 17020:2012** (Requirements for the operation of various types of bodies performing inspection)

A number of other standards and initiatives are evolving, including:

- The **ISO 27000** series of standards on information security management also contains a standard **ISO 27037:2012** (Guidelines for identification, collection, acquisition, and preservation of digital evidence)
- US Department of Justice Computer Crime and Intellectual Property Section provides a manual on Searching and Seizing Computers and Obtaining Electronic Evidence in Criminal Investigations
- The Scientific Working Group on Digital Evidence (SWGDE) is a long-standing effort offering inter alia the SWGDE Best Practices for Computer Forensics

Digital Evidence

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Digital Evidence

"If you want to learn to solve problems, you must be ready to look anywhere, for anything, and you must be prepared when you find it."
(Farmer and Venema, 2004)

- Digital evidence is any kind of digitally processed information that is stored in any sort of digital media.
  - Strengthens or negates the assumption of an electronic crime in the terms of the investigation process
  - can be presented as supportive proof in a court of law.

Advantages of Digital Evidence

- Can have exact duplicates
- The investigator can work with a copy and preserve the original
- The right tools can determine tampered digital evidence
- Fairly difficult to get destroyed
- Evidence can exist there where criminals cannot be aware of.
- It can be examined later in a controlled environment.
- The risk of damaging and/or disrupting business services is minimum

Disadvantages of Digital Evidence

- Risk of destroying/ missing evidence
- Can exist for a fraction of a second.
- Requires equipment and technical expertise
- The collection method may significantly alter the data.
- Risk not being able to access all evidence on a drive, e.g., encrypted data
- Time consuming
- Methods are open to critisism

Evidence and Practices

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Evidence types

"Electronic evidence is valuable evidence, and it should be treated in the same manner as traditional forensic evidence - with respect and care."

ACPO

Inculpatory

Verifies existing data and theories

Exculpatory

Contradicts existing data and theories

Evidence Categories

Circumstantial Evidence	- draws an inference of guilt sufficiently to prove a case beyond reasonable doubt; - it is not drawn from fact and can be open to interpretation.
Direct Evidence	- comprises of evidence of the allegation itself - if that evidence is accepted then the allegation itself will be proven.

Digital Evidence Guidelines

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Admissible

The evidence must conform to certain legal rules before it can be put before a court.

Authentic

It must be possible to positively tie the evidentiary material to the incident by showing that the evidence relates to the incident in a relevant way.

Complete

It must tell the whole story and not just one particular perspective. Not only should you collect evidence that can prove the attacker's actions, but also evidence that could prove his or her innocence.

Reliable

There must be nothing about how the evidence was collected and subsequently handled that casts doubt about its authenticity and veracity.

Believable and understandable

The evidence must be readily believable and understandable by a court of law.

Providing or Giving Evidence

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Evidence may be provided in different forms:

- **Witness Evidence**: This can be provided in several forms either from eyewitnesses or expert witnesses drawing on specialist skills. Transcripts of interviews under caution can be used, recording is used under the Police and Criminal Evidence Act 1984 (PACE) except normally in Scotland
- **Documentary Evidence**: This is any evidence introduced in the form of documents and is not limited to written documents; this is commonly subject to some form of authentication
- **Real Evidence**: Physical evidence introduced in the form of material objects; this is intended to prove a fact based on demonstrable and verifiable physical characteristics

Forms of Evidence

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Real Evidence

- an object that is involved in a case or may have played a role in the action in question.
- The object may also be able to be examined by a Court or an expert witness who is suitably qualified to provide guidance to the Court regarding the item.

Testimonial Evidence

- provided by a witness
- This may be a witness to the alleged offence itself or a professional witness who has examined the item of real evidence and are providing findings based on that examination.

Hearsay Evidence

- less common
- information given to a Court by someone who is not a direct witness of an event or action and does not have first-hand knowledge of something but has heard it from another source.

Sources of Digital Evidence

Any digital device

- Hard drives
- Memory
- Removable storage
- Tablets
- Mobile phones
- Network devices
- Printers
- Scanners
- Dongles
- Smart Cards
- Biometric Scanners
- Virtual desktops

Examples of digital evidence

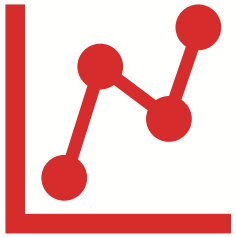
- System logs
- Internet activity logs
- IP addresses
- Tools
- Configuration files
- E-mails, notes
- Text files
- Malicious software
- Source code
- Images
- Address books

Any **digital file/ source**

The digital evidence problems

Complexity Problem

1. acquired data are typically at the lowest and most raw format
2. difficult for investigators to understand
3. solved by using tools to translate data



Quantity Problem

1. the amount of data to analyse can be very large
2. data reduction techniques are utilised

Determining What Data to Collect and Analyse

- Examining and analyzing digital evidence depend on the nature of the investigation
  - And the amount of data to process
- **Scope creep** - when an investigation expands beyond the original description
  - Because of unexpected evidence found
  - You may be asked investigators to examine other areas to recover more evidence
  - Increases the time and resources needed to extract, analyse, and present evidence
  - Criminal investigations require more detailed examination of evidence just before trial

Summary

- Cybercrimes are cyber-enabled or cyber-dependent
- The role of digital forensics in different types of investigations
  - Identify the different specialist areas within digital forensics
- Digital forensics can be used for all types of different cases
- Evidence can be circumstantial or direct

Questions?

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Discussion point

Discuss the benefits and shortcomings of creating specialisations in digital forensics

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Appendix

Additional notes

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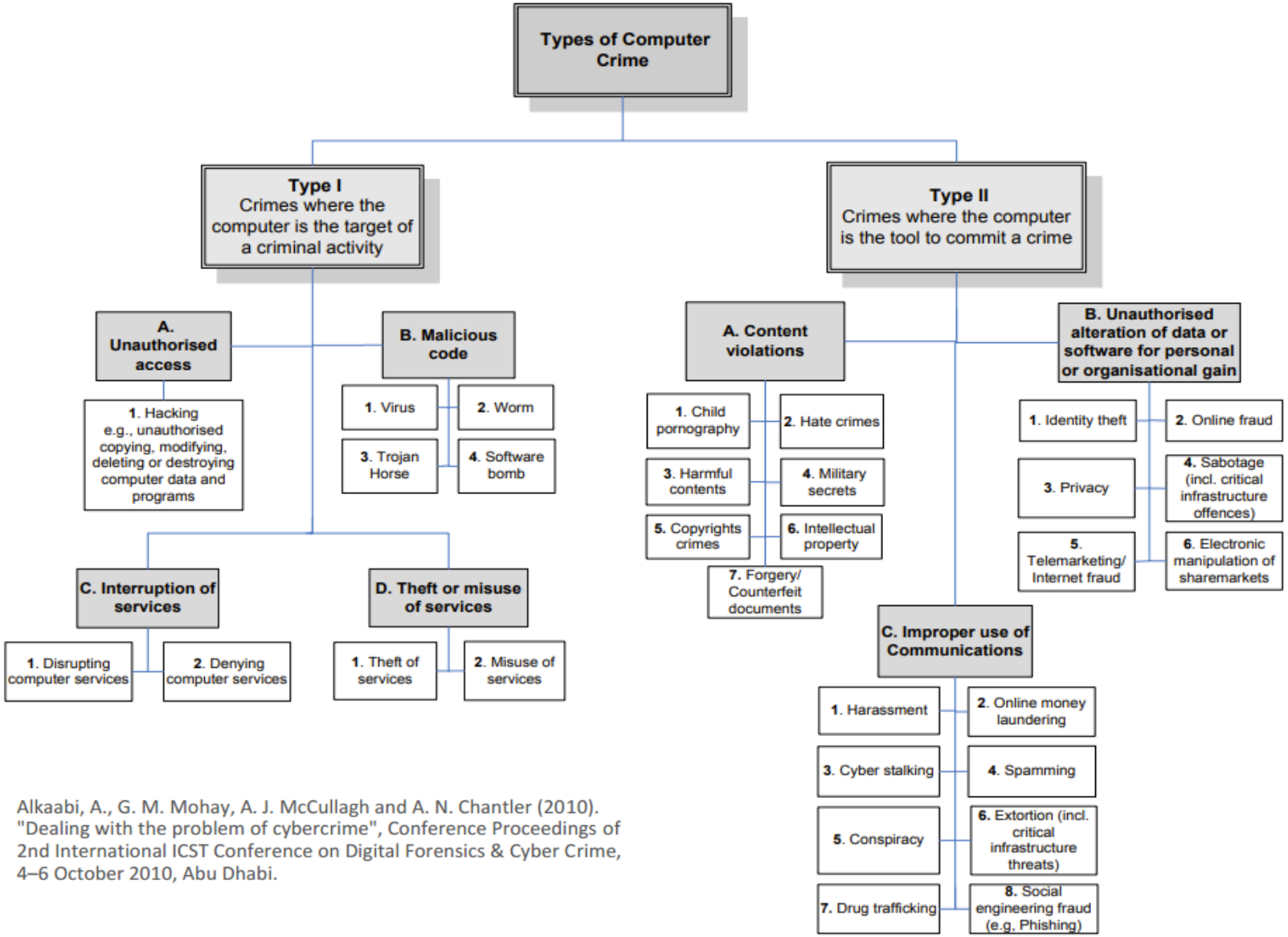

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